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MARI — Detailed Weekly Report

Week 2: Theory — Cost Bounds, Optimality & Coupling

Member A (Theory & Writing) • Member B (Implementation & Evaluation)

1. Objective

Prove the mechanism's cost bounds and the result that separates MARI from ordered indexes, and validate them empirically.

2. Motivation — why this week

A working mechanism is not yet a contribution; a referee asks whether the cost is provably bounded and what fundamentally distinguishes MARI. We supply both, and we report a negative result honestly rather than overclaiming.

3. Method & key equations

$$\text{Compaction (Thm 1): compact when } |D_j| \geq \varepsilon|S_j| \Rightarrow O(1/\varepsilon) \text{ amortized; space } O(n+m).$$
$$\text{Maintenance (Thm 3): adaptive split/merge is } O(1) \text{ amortized.}$$
$$\text{Cor. 4: } \Pr[\text{migrate}] \leq \Pr[|\Delta k| > g].$$
$$\text{Prop. 2 (coupling): single width } W \Rightarrow r(W)\downarrow,\ s(W)\uparrow;\ \text{MARI decouples } g \text{ from } w.$$

4. Process (step by step)

1. Prove Theorems 1–3, Corollaries 1–4, Proposition 1 (with its refutation), and Proposition 2 (relocation–scan coupling).
2. Implement adaptive split/merge (`mari_adaptive.py`); sanity-check with `adaptive_test.py`.
3. Run `python thm_check.py` to measure rebuild-work per update against the $O(1)$ prediction.
4. Run `lb_check.py/lb_frontier.py` to test the tempting (refuted) lower bound.

5. Results

Claim	Statement	Check
Theorem 1	compaction $O(1/\varepsilon)$, space $O(n+m)$	matches sweep
Theorem 3	maintenance $O(1)$ amortized	≈0.27/update
Prop. 2	relocation–scan coupling	separates MARI

- Theorem 3 validated at ≈**0.27** rebuild-work per update, 0 mismatches.
- Proposition 1's tempting lower bound is *refuted* and published as a negative result.
- Proposition 2 explains why every ordered competitor is pinned to ≈1.0 relocations.

6. Interpretation

The bounds make MARI predictable rather than heuristic, and the coupling result gives the structural reason (decoupling the guard g from the scan width w) that competitors cannot avoid relocation.

7. Who did what

- **Member A:** Proved all theorems/propositions and wrote the theory section, including the honest refutation.
- **Member B:** Built the validation harness and the adaptive split/merge implementation; ran the lower-bound checks.

8. Deliverables (code)

`thm_check.py`, `mari_adaptive.py`, `adaptive_test.py`, `lb_check.py`, `lb_frontier.py`

9. Status

Status: Completed.

10. Next week

Week 3 tests the theory empirically against real ordered-index competitors.